Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Models | Diffusion Transformers

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Website | LinkedIn | GitHub | Google Scholar

Postdoctoral researcher with a PhD in Artificial Intelligence and Robotics. Develops multimodal generative models for human motion and spatial audio, combining diffusion transformers, representation learning, dataset development, and experimental evaluation in Python and PyTorch.

PROFESSIONAL EXPERIENCE

ITALIAN INSTITUTE OF TECHNOLOGY (IIT) | Postdoctoral Researcher

Feb 2025 - Present | Genoa, Italy

Multimodal learning and spatial audio for immersive environments in the European XTREME project.

- Developing few-shot room impulse response generation from audio and depth, with counterfactual training for reverberation control (ongoing research).
- Built Eigenmike 64 processing pipelines for beamforming, binaural rendering, and acoustic mapping on 3D room point clouds.
- Implemented audio/video synchronization and 360-degree acoustic overlays for spatial-audio recordings.

UNIVERSITY OF GENOA | PhD Researcher

Nov 2021 - May 2025 | Genoa, Italy / London, UK

Research on multimodal human motion generation and domain generalization, including a research collaboration with King's College London.

- Designed ALaDiT, a diffusion transformer conditioned on speech, text, and cultural representations for 3D gesture generation (SICAGE, ECCV 2026).
- Built a video-to-3D-motion pipeline and released TED4C-L: 106.45 hours of aligned motion, audio, and text from 764 speakers.
- Learned cultural representations with Fishr and adversarial domain generalization, using speaker-disjoint training and evaluation.
- Evaluated motion quality, diversity, and speech alignment through baseline comparisons and controlled ablations; released training and evaluation code.
- Developed semantic-similarity algorithms for symbolic and deictic gesture labeling (IEEE RO-MAN 2024).

AKKA TECHNOLOGIES | Software Engineer

May 2021 - Oct 2021 | Italy

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

RESEARCH FOCUS AND TECHNICAL SKILLS

Generative Modeling: Diffusion transformers; GANs; Multimodal conditioning.

Learning and Evaluation: Representation learning; Domain generalization; Generative model evaluation.

Programming: Python; C/C++; MATLAB/Simulink.

ML Frameworks: PyTorch; Scikit-learn; TensorFlow.

Tools and Training: Git; Linux; SLURM; Data parallelism.

EDUCATION

UNIVERSITY OF GENOA | Ph.D. in Artificial Intelligence and Robotics

Nov 2021 - May 2025

Research collaboration with King's College London; thesis defended in May 2025.

UNIVERSITY OF GENOA | M.Sc. in Robotics Engineering

2018 - 2021

Thesis: Culture-Aware Co-Speech Gestures Using Generative Adversarial Networks.

Developed speech-driven GAN gesture generation tested on the Pepper robot (AIRO 2021; International Journal of Social Robotics 2022).

UNIVERSITY OF GENOA | B.Sc. in Biomedical Engineering

2014 - 2018

PUBLICATIONS

A. Gjaci, A. Sgorbissa, V. Murino. SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. European Conference on Computer Vision (ECCV), main conference, pp. 392-411. 2026.

A. Gjaci, V. Schmuck, A. Sgorbissa, O. Celiktutan. Exploring Cultural Cues in Multimodal Interactions. AHRI Workshop at ACII, pp. 306-313. 2024.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, pp. 477-484. 2024.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. Towards Culture-Aware Co-Speech Gestures for Social Robots. International Journal of Social Robotics, 14(6), pp. 1493-1506. 2022.

A. Gjaci, L. Oneto, C. T. Recchiuto, A. Sgorbissa. Culture Awareness in Intelligent Systems. AIRO Workshop. 2022.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. A GAN-based Approach for Generating Culture-Aware Co-Speech Gestures. AIRO Workshop. 2021.

TEACHING AND SERVICE

- Teaching assistant: Foundations of Informatics (C++), 60 hours; Mobile Programming (Android/Kotlin), 30 hours.
- Co-organizer of the ECCV workshop Expressive Encounters.
- Reviewer for AAAI, ECCV, ACM Multimedia, IROS, ICMI, and the International Journal of Social Robotics.
- Session chair at IEEE RO-MAN 2024.

SELECTED TRAINING

Topics in Modern Machine Learning (MaLGa, 2023); Robotics and Intelligent Machines Summer School (Scuola Superiore Sant'Anna, 2023).

LANGUAGES

Italian (native); Albanian (native); English (fluent).